

ISTA and LASSO: Math 535 Project

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April 14, 2025

Why We Chose ISTA as Our Project Topic

- **Solves LASSO Efficiently**

LASSO is widely used in feature selection, sparse regression, and high-dimensional statistics. ISTA provides an elegant, iterative solution.

- **Teachable, Intuitive, and Theoretically Rich**

The algorithm is simple to understand, easy to implement, and its convergence properties can be fully analyzed using convex analysis tools.

- **Practical Relevance**

ISTA and its accelerated variants (like FISTA) are widely used in modern machine learning libraries and scientific computing.

- **Aligned with Math 535**

This project connects to core topics in Math 535 such as gradient methods, Lipschitz continuity, and composite convex optimization.

ℓ_0 vs ℓ_1 Norms

ℓ_0 norm:

$\|\beta\|_0 = \text{Number of non-zero elements in } \beta$

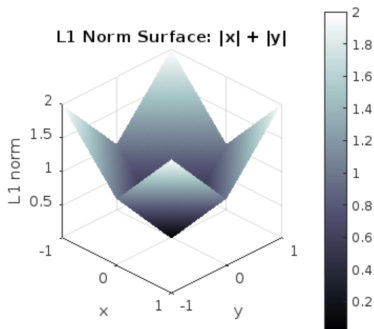
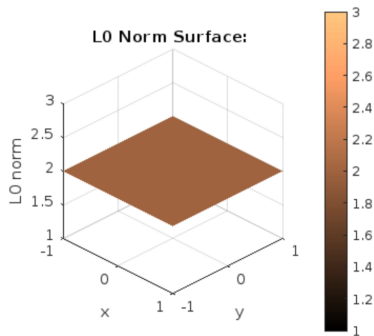
ℓ_1 norm:

$$\|\beta\|_1 = \sum_{i=1}^n |\beta_i|$$

- ℓ_0 : Discrete, nonconvex, and hard to solve through the standard optimization algorithm (NP-hard).
- ℓ_1 : Continuous, convex, promotes sparsity.

Geometric View: ℓ_0 vs ℓ_1

- ℓ_0 norm level sets: sparse grid-like, non-convex jumps
- ℓ_1 norm level sets: diamond shape in 2D — convex



Proof: ℓ_0 Norm is not Convex and not differentiable

1. Not Convex: Convexity requires:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \lambda \in [0, 1]$$

Counterexample:

$$\mathbf{x} = [1, 0], \quad \mathbf{y} = [0, 1], \quad \|\mathbf{x}\|_0 = \|\mathbf{y}\|_0 = 1$$

$$\mathbf{z} = 0.5\mathbf{x} + 0.5\mathbf{y} = [0.5, 0.5] \Rightarrow \|\mathbf{z}\|_0 = 2$$

$$0.5\|\mathbf{x}\|_0 + 0.5\|\mathbf{y}\|_0 = 1 < 2$$

Therefore, the convexity inequality fails. $\Rightarrow \ell_0$ norm is **not convex**.

2. Not Differentiable: A function is differentiable if it is continuous and has a defined derivative.

But for ℓ_0 : If x_i changes from the small value $0 \rightarrow$, $\|\mathbf{x}\|_0$ increases by 1. If x_i changes from the small value $\rightarrow 0$, $\|\mathbf{x}\|_0$ drops by 1. $\Rightarrow$ No well-defined tangent at such points. $\Rightarrow \ell_0$ norm is **not differentiable**.

Proof: ℓ_1 Norm is Convex

Definition of ℓ_1 norm:

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

To prove: $f(x) = \|x\|_1$ is convex, i.e.,

$$\|\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}\|_1 \leq \alpha \|\mathbf{x}\|_1 + (1 - \alpha) \|\mathbf{y}\|_1, \quad \forall \alpha \in [0, 1]$$

Reasoning:

- The absolute value function $f_i(x) = |x|$ is convex on $\mathbb{R}$.
- Convexity holds even though $|x|$ is not differentiable at $x = 0$ (subgradients exist).
- The sum of convex functions is convex.

Therefore: $\|x\|_1 = \sum_i |x_i|$ is convex as a sum of convex functions.

LASSO Objective Function

We consider the following objective function for LASSO regression:

$$\min_{w \in \mathbb{R}^d} \left\{ \frac{1}{n} \sum_{i=1}^n (y_i - \langle w, x_i \rangle)^2 + \lambda \|w\|_1 \right\}$$

- $x_i \in \mathbb{R}^d$: input feature vectors
- $y_i \in \mathbb{R}$: target response
- $w \in \mathbb{R}^d$: parameter vector to be learned
- $\lambda > 0$: regularization parameter

Structure:

- $\frac{1}{n} \sum (y_i - \langle w, x_i \rangle)^2$ is the square loss — encourages data fidelity
- $\lambda \|w\|_1$ is the ℓ_1 penalty — encourages sparsity in w

Goal: Balance prediction accuracy and model simplicity by inducing sparsity in w .

Why Gradient Descent Fails for ℓ_1 Regularization

Background: The LASSO objective includes a smooth loss and a non-smooth regularization term.

Issue: ℓ_1 norm is not differentiable at zero.

$$\frac{d}{dx}|x| = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ \text{undefined} & x = 0 \end{cases}$$

$\Rightarrow$ Gradient descent cannot compute the update direction when $\beta_i = 0$

Why this matters:

- Gradient descent requires full differentiability of the objective.
- The non-differentiability of $\|\beta\|_1$ breaks the standard update rule.

Theme: Proximal Gradient Methods

- Split the objective: smooth part + non-smooth convex part
- Use a gradient step for the smooth part
- Use a **proximal operator** for the non-smooth part ($\rightarrow$ soft-thresholding)¹

¹See Tibshirani, R. J. (2015), *Proximal Gradient Descent and Acceleration*.

Why QR Decomposition Doesn't Work for LASSO

- **Normal equations:** $X^T X \beta = X^T y$ assume differentiable convex objective
- ℓ_1 term is not differentiable, objective is convex but non-smooth $\Rightarrow$ no closed-form solution
- **QR decomposition** assumes same conditions as normal equations
- Instead: use **ISTA** (iterative soft-thresholding)

Why Do We Need the Gradient of Least Squares?

- Back to the Objective Function.
- We now write LASSO objective in the following form: $F(w) + R(w)$ where:
 - $F(w) = \frac{1}{n} \sum_{i=1}^n (y_i - \langle w, x_i \rangle)^2$ (smooth)
 - $R(w) = \lambda \|w\|_1$ (non-smooth)
- Proximal gradient methods (like ISTA) require computing $\nabla F(w)$
- Gradient of least squares:

$$\nabla F(w) = -\frac{2}{n} X^T (y - Xw)$$

- Needed to take a gradient step before applying the soft-thresholding operator

How Do We Derive ISTA?

- Goal: minimize $F(w) + R(w)$ using iterative updates
- At each iteration t :

$$w^{(t+1)} = \text{prox}_{\alpha R} \left(w^{(t)} - \alpha \nabla F(w^{(t)}) \right)$$

- For $R(w) = \lambda \|w\|_1$, the proximal operator becomes soft thresholding:

$$S_{\alpha\lambda}(z) = \text{sign}(z) \cdot \max(|z| - \alpha\lambda, 0)$$

- Final ISTA update rule:

$$w^{(t+1)} = S_{\alpha\lambda} \left(w^{(t)} - \alpha \nabla F(w^{(t)}) \right)$$

Convergence of ISTA (I): Assumptions and Theorem

We aim to minimize: $f(w) = F(w) + R(w)$

Assumptions:

- $F(w)$ is convex and differentiable over $\mathbb{R}^d$
- $\nabla F(w)$ is L -Lipschitz continuous:

$$\|\nabla F(w_1) - \nabla F(w_2)\| \leq L\|w_1 - w_2\|$$

- $R(w)$ is convex (possibly non-differentiable)

Theorem: If $\alpha \in (0, \frac{1}{L}]$, then ISTA satisfies:

$$f(w^{(k)}) - f(w^*) \leq \frac{\|w^{(0)} - w^*\|^2}{2\alpha k}$$

Convergence rate: $\mathcal{O}(1/k)$

Convergence of ISTA (II): Proof Sketch – Part 1

Step 1: Gradient update

$$\begin{aligned}w_{k+1} &= w_k - \alpha \nabla F(w_k) \\ \Rightarrow \|w_{k+1} - w^*\|^2 &= \|w_k - \alpha \nabla F(w_k) - w^*\|^2 \\ &= \|w_k - w^*\|^2 + \alpha^2 \|\nabla F(w_k)\|^2 - 2\alpha \nabla F(w_k)^T (w_k - w^*)\end{aligned}$$

Step 2: Convexity inequality (first-order condition)

$$\nabla F(w_k)^T (w_k - w^*) \geq F(w_k) - F(w^*)$$

Step 3: Plug into Step 1

$$\|w_{k+1} - w^*\|^2 \leq \|w_k - w^*\|^2 + \alpha^2 \|\nabla F(w_k)\|^2 - 2\alpha [F(w_k) - F(w^*)]$$

Step 4: Use descent lemma (L-smoothness)

$$\begin{aligned}\frac{1}{2\alpha} \|\nabla F(w_k)\|^2 &\leq F(w_k) - F(w_{k+1}) \\ \Rightarrow \|w_{k+1} - w^*\|^2 &\leq \|w_k - w^*\|^2 + 2\alpha [F(w^*) - F(w_{k+1})]\end{aligned}$$

Convergence of ISTA (III): Proof Sketch – Part 2

Step 5: Rearranging inequality

$$F(w_{k+1}) - F(w^*) \leq \frac{\|w_k - w^*\|^2 - \|w_{k+1} - w^*\|^2}{2\alpha}$$

Step 6: Sum over $k = 0$ to $N - 1$

$$\sum_{k=0}^{N-1} [F(w_{k+1}) - F(w^*)] \leq \frac{\|w_0 - w^*\|^2 - \|w_N - w^*\|^2}{2\alpha} \leq \frac{\|w_0 - w^*\|^2}{2\alpha}$$

Step 7: Use monotonicity $F(w_{k+1}) \leq F(w_k)$

$$F(w_N) - F(w^*) \leq \frac{1}{N} \sum_{k=0}^{N-1} [F(w_{k+1}) - F(w^*)] \leq \frac{\|w_0 - w^*\|^2}{2\alpha N}$$

Conclusion: ISTA converges at rate $\mathcal{O}(1/N)^2$

²See Beck and Teboulle (2009), A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems.

Is the LASSO / ISTA Solution Unique?

- LASSO solution may not be unique due to the ℓ_1 norm
- ℓ_1 norm is convex but not strictly convex
- Uniqueness depends on $F(w)$:
 - If $X^T X$ is positive definite, $F(w)$ is strictly convex
 - But overall objective is not strictly convex $\Rightarrow$ multiple solutions possible
- ISTA will converge to a minimizer, but not necessarily the unique one³

³See Tibshirani, R. J. (2012), *The Lasso Problem and Uniqueness*.

References

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